* Digital Communication *

SALEH RAMADAN

"Computer"

"Sheet 2"

"Random Processes"

"Solution"

*** Problem 1:** B.W = 1 MHz

f_s = 1.5 * Nyquist-Rate

$L = 256$, $\mu = 255$

Solution

a) $SNR \simeq \dfrac{3L^2}{[\ln(1+\mu)]^2} = 6393.97$

$$SNR|_{dB} = 38.06 \text{ dB} \quad \#\#$$

b) $SNR_{Req.} \geq 48.06$ dB

· Case 1: $f_{s_1} = 1.5 * \text{Nyquist Rate}$ → $2 * B.W$

$$f_{s_1} = 1.5 * (2 * 1)$$

$$\boxed{f_{s_1} = 3 \text{ MHz}}, \quad L_1 = 256$$

$$\boxed{n_1 = 8 \text{ bits / symbol}}$$

· Case 2: $f_{s_2} = 1.2 * \text{Nyquist_Rate}$

$$f_{s_2} = 1.2 * (2 * 1)$$

$$\boxed{f_{s_2} = 2.4 \text{ MHz}}$$

$$SNR_{req} \geq 10^{\frac{48.06}{10}}, \quad \frac{3L^2}{[\ln(1+\mu)]^2} \geq 10^{\frac{48.06}{10}}$$

$$L \geq 809 \implies n_2 \geq 10$$

We need: $B.W_1 = B.W_2$

$$0.5 * n_1 * f_{s_1} = 0.5 * n_2 * f_{s_2} \implies \boxed{n_2 = 10}$$

$$\boxed{SNR|_2 \simeq 50 \text{ dB}}$$ ✳

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* Problem 2: $x(t) = A\cos(\omega_c t + \theta)$

$\theta \sim \mathcal{U}(-\pi, \pi)$

$$f_\theta(\theta) = \frac{1}{2\pi} \quad , \quad -\pi \leq \theta \leq \pi$$

Solution

$$R_X(t_1, t_2) = E(x(t_1)\,x(t_2))$$

$$= E\left[A^2\cos(\omega_c t_1 + \theta)\cos(\omega_c t_2 + \theta)\right]$$

$$= A^2 * E(g(\theta))$$

$$= A^2 * \int_{\theta=-\pi}^{\pi} g(\theta) f_\theta(\theta)\, d\theta$$

Remember:

$\cos(u)\cos(v) = \frac{1}{2}[\cos(u-v) + \cos(u+v)]$

$$= A^2 * \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos(\omega_c t_1 + \theta)\cos(\omega_c t_2 + \theta)\, d\theta$$

$$= \frac{A^2}{4\pi}\left[\int_{-\pi}^{\pi} \cos(\omega_c(t_1 - t_2))\, d\theta + \int_{-\pi}^{\pi} \cos(\omega_c(t_1 + t_2) + 2\theta)\, d\theta\right]$$

$\int_{-\pi}^{\pi} \cos(\omega_c(t_1+t_2)+2\theta)\, d\theta \rightarrow 0$

1)

$$R_X(t_1,t_2) = \frac{A^2}{2\pi} * \cos\left(w_c(t_1-t_2)\right) \int_{-\pi}^{\pi} d\theta$$

$$R_X(t_1,t_2) = \frac{A^2}{2}\cos\left(w_c(t_1-t_2)\right)$$

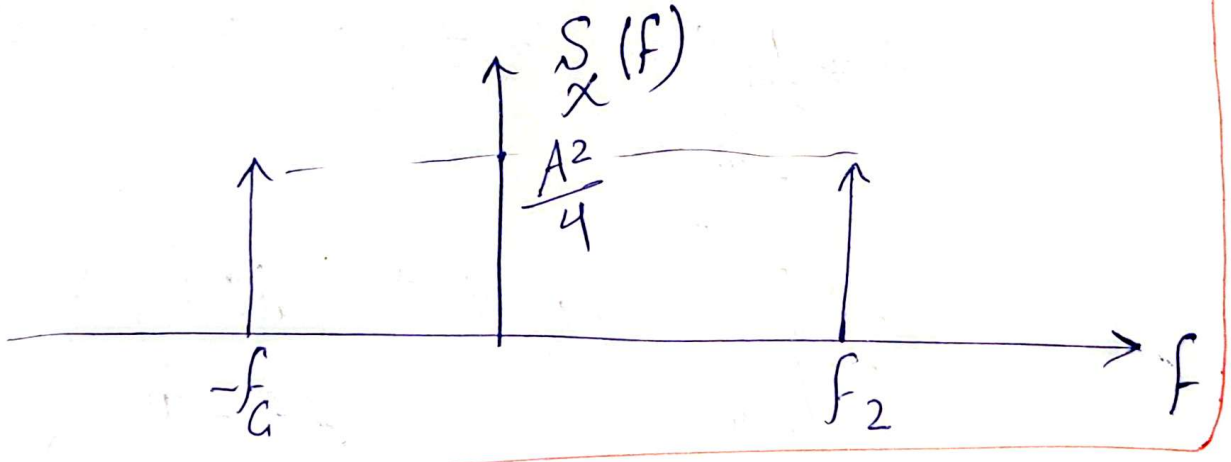
$$\boxed{R_X(\tau) = \frac{A^2}{2}\cos(w_c\tau)}$$

$$S_X(f) = F.T\left[R_X(\tau)\right]$$

~~$S_X(w) = \frac{A^2}{2} * \pi \left[\delta(w-w_c) + \delta(w+w_c)\right]$~~

$$S_X(w) = \frac{A^2}{2} * \pi * \left[\delta(w-w_c) + \delta(w+w_c)\right]$$

$$\boxed{S_X(f) = \frac{A^2}{4}\left[\delta(f-f_c) + \delta(f+f_c)\right]}$$



5)

* Problem 3:

$$y(t) = DC + g(t) + x(t)$$

$DC = \sqrt{\frac{3}{2}}$, $\mu_x = 0$, $g(t)$: Periodic Component , $\mu_g = 0$

Solution

$$R_Y(\tau) = E(y(t)\,y(t+\tau))$$

$$= E\left[\left(\sqrt{\tfrac{3}{2}} + g(t) + x(t)\right)\left(\sqrt{\tfrac{3}{2}} + g(t+\tau) + x(t+\tau)\right)\right]$$

$$= E\left[\tfrac{3}{2} + \sqrt{\tfrac{3}{2}}\,g(t+\tau) + \sqrt{\tfrac{3}{2}}\,x(t+\tau) + \sqrt{\tfrac{3}{2}}\,g(t)\right.$$

$$+ g(t)g(t+\tau) + g(t)x(t+\tau)$$

$$\left. + \sqrt{\tfrac{3}{2}}\,x(t) + x(t)\,g(t+\tau) + x(t)x(t+\tau)\right]$$

CamScanner

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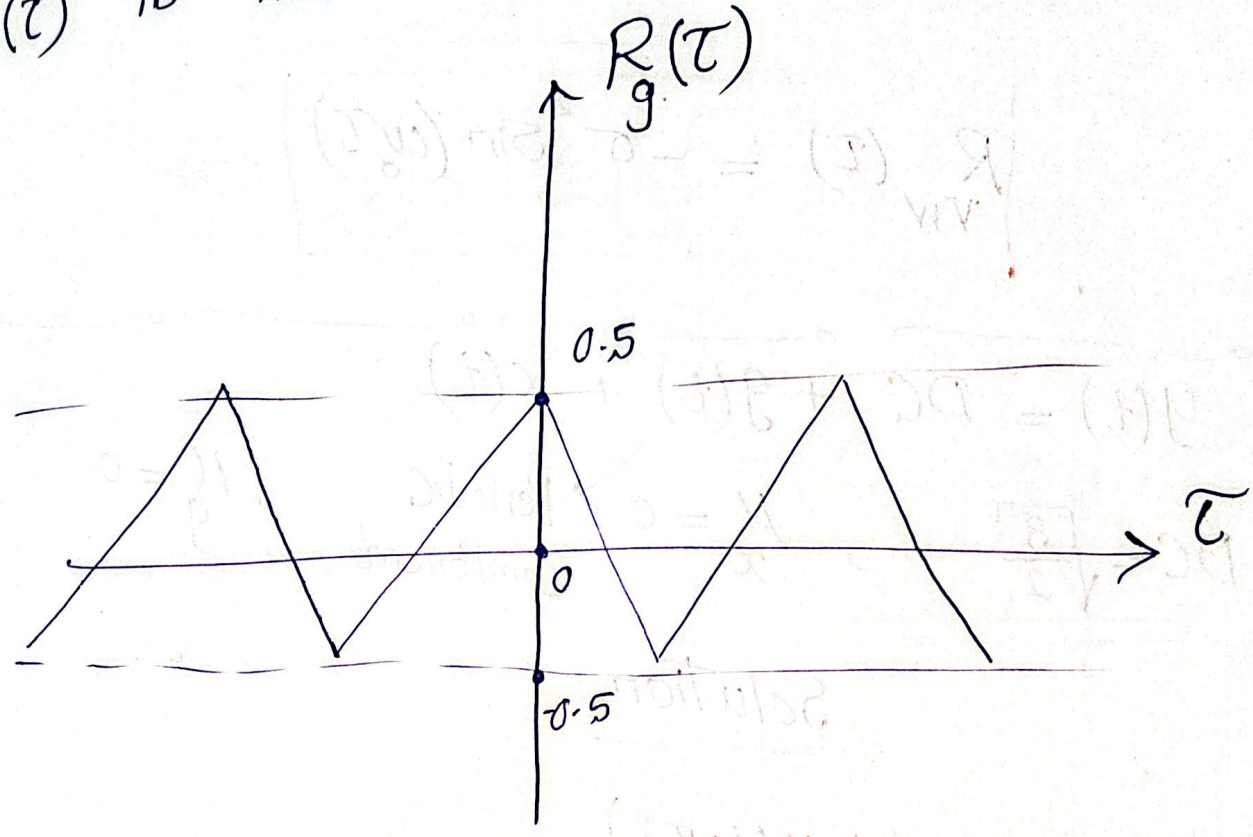
$$R_Y(\tau) = \frac{3}{2} + \sqrt{\frac{3}{2}}\,\mu_g * 2 + \sqrt{\frac{3}{2}} * \mu_x * 2 + 2\,\mu_g \mu_x + R_g(\tau) + R_x(\tau)$$

(μ_g and μ_x → 0)

$$R_Y(\tau) = \frac{3}{2} + R_g(\tau) + R_x(\tau)$$

Remove $\frac{3}{2}$ & $R_x(\tau)$ From Sketch($R_Y(\tau)$)

$R_x(\tau)$ is The APeriodiC Component



Average Power of $g(t) = R_g(0) = 0.5$

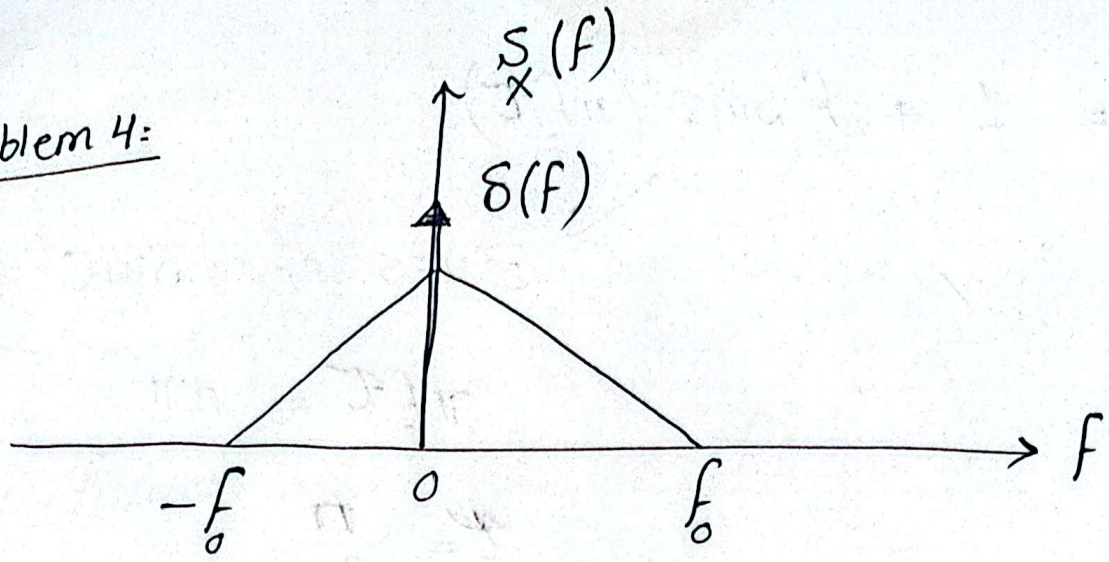
$$R_y(0) = \frac{3}{2} + R_g(0) + R_x(0) \Longrightarrow R_y(0) = 3$$

$$R_x(0) = 1 \Longrightarrow$$

Average Power of $g(t) = 0.5$
Average Power of $x(t) = 1$ ※

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***Problem 4:**

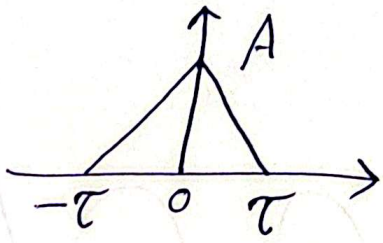


Solution

a) $S_x(f) = \delta(f) + \Delta\left(\frac{f}{2f_0}\right)$

$R_x(\tau) = I.F.T\left[S_x(f)\right]$

→ Remember: $A\ \Delta\left(\frac{t}{2\tau}\right)$



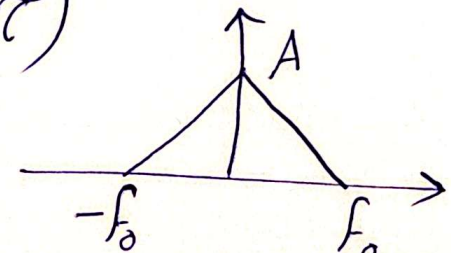
$\Downarrow$ F.T

$A\tau * Sinc^2(w * \frac{2\tau}{4})$

$A\tau * Sinc^2(2\pi f * \frac{2\tau}{4})$

$A\tau * Sinc^2(\pi f \tau)$

Reverse: $A * \Delta\left(\frac{f}{2f_0}\right)$



$\Downarrow$

$A * 2f_0 * Sinc^2(\pi t * f_0) \Longrightarrow A f_0\ Sinc^2(\pi f_0 t)$

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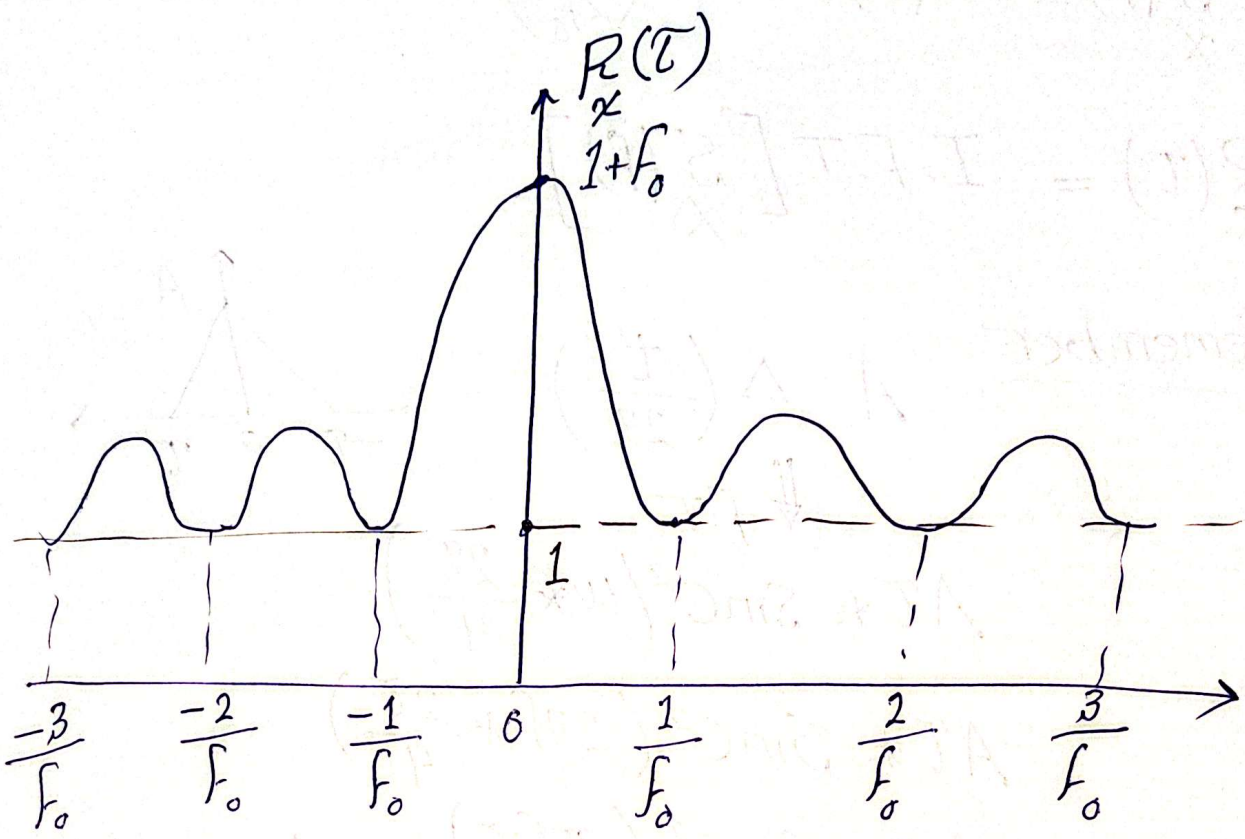
$$R_X(\tau) = 1 + f_0 \, \mathrm{sinc}^2(\pi f_0 \tau)$$

→ Zeros of sinc^2

$$\pi f_0 \tau = n\pi$$

$$\tau = \frac{n}{f_0}$$

$$\boxed{R_X(\tau) = 1 + f_0 \, \mathrm{sinc}^2(\pi f_0 \tau)} \quad //$$



b) DC. Power $= R_x(\infty)$

$$\boxed{\text{DC. Power} = 1}$$

$$\text{Total. Power} = R_x(0) = 1 + f_o$$

$$\text{A.C. Power} = \text{Total Power} - \text{DC. Power}$$

$$\boxed{\text{A.C. Power} = f_o}$$

c) If The samples of $x(t)$ is uncorrelated, This Means

$$Cov(x(t), x(t+\tau)) = 0$$

$$E(x(t), x(t+\tau)) = E(x(t))\,E(x(t+\tau))$$

$$R_x(\tau) = 1 * 1$$

$$R_x(\tau) = 1 \quad \text{at} \quad \tau = \frac{n}{f_o}$$

So These samples are uncorrelated